

Model Partial CC - 2022

- ① $a \in \mathbb{R}$? a.î. $d_1 \cap d_2$ și scrieți ec. planului determinat de cele două drepte.

$$d_1: \frac{x-2}{4} = \frac{y+1}{a} = \frac{z}{2} \Rightarrow \begin{cases} 2x-4=4z \\ 2y+2=az \end{cases} \Rightarrow \begin{cases} x-2=2z \\ 2y+2=az \end{cases}$$

$$d_2: \frac{x+1}{3} = \frac{y}{2} = \frac{z-2}{4} \Rightarrow \begin{cases} 2x+2=3y \\ 4y=2z-4 \end{cases} \Rightarrow \begin{cases} 2x+2=3y \\ 2y=z-2 \end{cases}$$

$$d_1 \cap d_2: \begin{cases} x-2=2z \rightarrow x=2+2z \\ 2y+2=az \\ 2x+2=3y \\ 2y=z-2 \end{cases} \rightarrow \begin{aligned} &4+4z+2=3y \\ &\Rightarrow 3y=4z+6 \\ &2y=z-2 \cdot 3 \Rightarrow 6y=3z-6 \\ &6y=8z+12 \Rightarrow 5z+18=0 \\ &z=-\frac{18}{5} \end{aligned}$$

$$6z+16+2=az \Rightarrow (a-6)z=18$$

$$\Rightarrow (a-6)\left(-\frac{18}{5}\right)=18 \Rightarrow 6-a=5 \Rightarrow \boxed{a=1}$$

$$v_1 \times v_2 = n = \begin{vmatrix} i & j & k \\ 4 & 1 & 2 \\ 3 & 2 & 4 \end{vmatrix} = 4j + 6k - 3k - 4i - 16j = -10j + 5k$$

$$x=0 \Rightarrow y = \frac{2}{3} \\ z = \frac{4}{3} + 2 = \frac{10}{3}$$

$$\pi: 0(x-0) + (-10)\left(y - \frac{2}{3}\right) + 5\left(z - \frac{10}{3}\right) = 0$$

$$\pi: -10y + \frac{20}{3} + 5z - \frac{50}{3} = 0 \Rightarrow \boxed{\pi: -10y + 5z - 10 = 0}$$

$$\textcircled{2} \quad A_1 = \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}_{\alpha_1} \quad A_2 = \begin{pmatrix} \beta & -3 \\ 2 & -1 \end{pmatrix} \quad A_3 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}_{\alpha_3} \quad A_4 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}_{\alpha_4}$$

a) A_1, A_2, A_3, A_4 linear dependent $\beta \in \mathbb{R}$?

$\downarrow$

$$A_2 = \alpha_1 A_1 + \alpha_3 A_3 + \alpha_4 A_4 \Rightarrow \begin{cases} \beta = 4\alpha_1 + \alpha_3 \\ -3 = 3\alpha_1 + \alpha_4 \\ 2 = 2\alpha_1 + \alpha_3 \\ -1 = \alpha_1 + \alpha_4 \end{cases} \rightarrow$$

$$\Rightarrow -1 + 3 = \alpha_1 - 3\alpha_1 \Rightarrow 2 = -2\alpha_1 \Rightarrow \underline{\alpha_1 = -1}$$

$$\alpha_4 = -1 - \alpha_1 = -1 - (-1) \Rightarrow \underline{\alpha_4 = 0}$$

$$2 = 2\alpha_1 + \alpha_3 \Rightarrow \alpha_3 = 2 - 2\alpha_1 = 2 - 2(-1) = 2 + 2 \Rightarrow \alpha_3 = 4$$

$$2 = 2\alpha_1 + \alpha_3 \Rightarrow \alpha_3 = 2 - 2\alpha_1 = 2 - 2(-1) = 2 + 2 \Rightarrow \underline{\alpha_3 = 4}$$

$$\beta = 4 \cdot (-1) + 4 = 0 \Rightarrow \boxed{\beta = 0}$$

$$A_1; A_2; A_3; A_4 \text{ lin. dep.} \Rightarrow \begin{pmatrix} 4 & \beta & 1 & 0 \\ 3 & -3 & 0 & 1 \\ 2 & 2 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix} \xrightarrow{L_1 \leftrightarrow L_4}$$

$$\begin{pmatrix} 1 & -1 & 0 & 1 \\ 3 & -3 & 0 & 1 \\ 2 & 2 & 1 & 0 \\ 4 & \beta & 1 & 0 \end{pmatrix} \xrightarrow{\begin{matrix} L_2 - 3L_1 \\ L_3 - 2L_1 \\ L_4 - 4L_1 \end{matrix}} \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & -2 \\ 0 & 4 & 1 & -2 \\ 0 & \beta + 4 & 1 & -4 \end{pmatrix} \xrightarrow{L_4 \leftrightarrow L_2}$$

$$\begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & \beta + 4 & 1 & -4 \\ 0 & 4 & 1 & -2 \\ 0 & 0 & 0 & -2 \end{pmatrix} \xrightarrow{C_2 \leftrightarrow C_3} \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & \beta + 4 & -4 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & -2 \end{pmatrix} \xrightarrow{L_2 \leftrightarrow L_3}$$

$$\begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 4 & -2 \\ 0 & 1 & \beta+4 & -4 \\ 0 & 0 & 0 & -2 \end{pmatrix} \xrightarrow{L_3-L_2} \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & \beta & -2 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

$$\text{lin. ind} \Leftrightarrow \boxed{\beta \neq 0}$$

$$\Rightarrow \text{lin. dep. pt. } \boxed{\beta = 0}$$

b) Pt β determinat anterior completăm o bază a subspațiului $\text{Sp}\{A_1, A_2, A_3, A_4\}$ la o bază a lui $M_2(\mathbb{R})$ și determinăm coordonatele matricei $B = \begin{pmatrix} 5 & 4 \\ 3 & 2 \end{pmatrix}$ în această bază.

$$\beta = 0 \Rightarrow A_1 = \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix} \quad A_2 = \begin{pmatrix} 0 & -3 \\ 2 & -1 \end{pmatrix} \quad A_3 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \quad A_4 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\text{Sp}\{A_1, A_2, A_3, A_4\} = \{A_1, A_3, A_4\}$$

$$\text{luăm } P = \begin{pmatrix} 1 & -3 \\ 2 & -1 \end{pmatrix}$$

$$\Rightarrow \text{atunci } \Rightarrow \text{cf. a)} \Rightarrow \text{Sp}\{A_1, P, A_2, A_3\}$$

$$\text{din } \text{Sp}\{A_1, P, A_2, A_3\} = 4 \text{ formează bază}$$

$$\left(\begin{array}{cccc|cccc} 4 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 3 & -3 & 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow{L_1 \leftrightarrow L_4}$$

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 3 & -3 & 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 0 & 1 & 0 \\ 4 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{array} \right) \xrightarrow{\begin{array}{l} L_2 - 3L_1 \\ L_3 - 2L_1 \\ L_4 - 4L_1 \end{array}}$$

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 & 0 & 1 & 0 & -3 \\ 0 & 4 & 1 & -2 & 0 & 0 & 1 & -2 \\ 0 & 5 & 1 & -4 & 1 & 0 & 0 & -4 \end{array} \right) \xrightarrow{L_2 \leftrightarrow L_4}$$

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 5 & 1 & -4 & 1 & 0 & 0 & -4 \\ 0 & 4 & 1 & -2 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & -2 & 0 & 1 & 0 & -3 \end{array}\right) \xrightarrow{L_2/5}$$

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1/5 & -4/5 & 1/5 & 0 & 0 & -4/5 \\ 0 & 4 & 1 & -2 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & -2 & 0 & 1 & 0 & -3 \end{array}\right) \xrightarrow{L_3 - 4L_2}$$

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1/5 & -4/5 & 1/5 & 0 & 0 & -4/5 \\ 0 & 0 & 1/5 & 8/5 & -4/5 & 0 & 1 & 6/5 \\ 0 & 0 & 0 & -2 & 0 & 1 & 0 & -3 \end{array}\right) \xrightarrow{5 \cdot L_3}$$

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1/5 & -4/5 & 1/5 & 0 & 0 & -4/5 \\ 0 & 0 & 1 & 6 & -4 & 0 & 5 & 6 \\ 0 & 0 & 0 & -2 & 0 & 1 & 0 & -3 \end{array}\right) \xrightarrow{L_4/(-2)}$$

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1/5 & -4/5 & 1/5 & 0 & 0 & -4/5 \\ 0 & 0 & 1 & 6 & -4 & 0 & 5 & 6 \\ 0 & 0 & 0 & 1 & 0 & -1/2 & 0 & 3/2 \end{array}\right) \xrightarrow{L_3 - 6L_4}$$

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1/5 & -4/5 & 1/5 & 0 & 0 & -4/5 \\ 0 & 0 & 1 & 0 & -4 & 3 & 5 & -3 \\ 0 & 0 & 0 & 1 & 0 & -1/2 & 0 & 3/2 \end{array}\right) \begin{array}{l} \xrightarrow{L_2 - \frac{1}{5}L_3} \\ \xrightarrow{L_2 + \frac{4}{5}L_4} \end{array} \dots$$

$$\textcircled{3} f: \mathbb{R}^3 \longrightarrow \mathbb{R}^3 \quad f\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} \delta x + y + z \\ x + \delta y + z \\ x + y + \delta z \end{pmatrix}$$

a) Determinați $M(f)$ în raport cu baza canonică;
 $\text{Ker}(f)$, $\text{Im}(f)$ (cu o bază și dimensiunea)

$$M(f)_{cc} = \begin{pmatrix} \delta & 1 & 1 \\ 1 & \delta & 1 \\ 1 & 1 & \delta \end{pmatrix}$$

$$\text{Ker}(f) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid f\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = 0 \right\} \Rightarrow \begin{cases} \delta x + y + z = 0 \\ x + \delta y + z = 0 \\ x + y + \delta z = 0 \end{cases}$$

$$\begin{vmatrix} \delta & 1 & 1 \\ 1 & \delta & 1 \\ 1 & 1 & \delta \end{vmatrix} = \delta^3 + 1 + 1 - \delta - \delta - \delta = \delta^3 - 3\delta + 2$$

$$\Delta = (\delta - 1)^2(\delta + 2)$$

	δ^3	δ^2	δ	δ^0
	1	0	-3	2
δ	1	1	-2	0
δ^2	1	1	2	0

I $\Delta \neq 0; \delta \in \mathbb{R} \setminus \{1, -2\} \Rightarrow \text{rang } M(f) = 3 \Rightarrow x = y = z = 0$
 $\Rightarrow \underline{\text{Ker}(f) = \{0\}} \Rightarrow \underline{\dim \text{Ker}(f) = 0}$ $\xrightarrow[\mathbb{R}-\text{D}]{\text{Th.}}$ $\dim_{\mathbb{K}} \text{Im}(f) = 3$
 $\dim \mathbb{R}^3 = 3$ $\Rightarrow \underline{\text{Im}(f) = \mathbb{R}^3}$

$\Rightarrow$ o bază este $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

II $\Delta = 0$ pt. că $\delta = -2 \Rightarrow \text{rang } M(f) = 2$

Notăm $z = \alpha$ nec. nec. $\Rightarrow \begin{cases} -2x + y = -\alpha \Rightarrow y = 2x - \alpha \\ x - 2y = -\alpha \end{cases}$

$\Rightarrow x - 2(2x - \alpha) = -\alpha \Rightarrow x - 4x + 2\alpha = -\alpha \Rightarrow$

$$\Rightarrow -3x = -3\alpha \Rightarrow x = \alpha \quad y = 2\alpha - \alpha = \alpha \quad \text{pt } \ker(f) = \text{Sp}\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$$

$$\underline{\dim \ker(f) = 1}$$

Th. $\frac{R-D}{R-D}$, $\dim \text{Im}(f) = 3 - 1 \Rightarrow \underline{\dim \text{Im}(f) = 2}$

$$\text{Im}(f) = \left\{ f\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) \mid \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \right\} = \left\{ \begin{pmatrix} -2x + y + z \\ x - 2y + z \\ x + y - 2z \end{pmatrix} \mid \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \right\}$$

$$\text{Im}(f) = \text{Sp}\left\{\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}\right\}$$

$$\dim \text{Im}(f) = 2 \Rightarrow \underline{\text{Im}(f) = \text{Sp}\left\{\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}\right\}}$$

o bază este $\left\{\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}\right\}$

III) $\Delta = 0$ pt. c.ă $\delta = 1 \Rightarrow \text{rang } M(f) = 1$

Notăm $y = \alpha$ nec. nec. $\Rightarrow x = -\alpha - \beta \Rightarrow$
 $z = \beta$

$$\Rightarrow \ker(f) = \text{Sp}\left\{\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right\}$$

$$\underline{\dim \ker(f) = 2}$$

Th. $\frac{R-D}{R-D}$, $\dim \text{Im}(f) = 3 - 2 \Rightarrow \underline{\dim \text{Im}(f) = 1}$

$$\text{Im}(f) = \left\{ f\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) \mid \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \right\} = \left\{ \begin{pmatrix} x + y + z \\ x + y + z \\ x + y + z \end{pmatrix} \mid \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \right\}$$

$$\Rightarrow \underline{\text{Im}(f) = \text{Sp}\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}} \Rightarrow \text{o bază este } \left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$$

b) $\delta = ?$ a.î. suma dintre $\ker(f)$ și $\text{Im}(f)$
 să fie directă

$$\text{Sumă directă} \Rightarrow \ker(f) \cap \operatorname{Im}(f) = \{0\}$$

$\Downarrow$

$$\dim \ker(f) + \dim \operatorname{Im}(f) = \dim [\ker(f) \oplus \operatorname{Im}(f)]$$

Conform cazurilor de la a avem:

$$\textcircled{\text{I}} \text{ pt. } g \in \mathbb{R} \setminus \{1, -2\} \Rightarrow \begin{array}{l} \ker(f) = \{0\} \\ \operatorname{Im}(f) = \mathbb{R}^3 \end{array} \not\Rightarrow$$

$$\Rightarrow \text{se respectă } \ker(f) \cap \operatorname{Im}(f) = \{0\}$$

$$\Rightarrow \text{e sumă directă pt. } g \in \mathbb{R} \setminus \{1, -2\}$$

II

④ $A = \begin{pmatrix} 2 & 1 & -2 \\ 2 & 3 & -4 \\ 1 & 1 & -1 \end{pmatrix}$ Să se verifice dacă este diagonalizabilă, și în caz afirmativ să se aducă la forma diagonală.

$$\begin{vmatrix} 2-\lambda & 1 & -2 \\ 2 & 3-\lambda & -4 \\ 1 & 1 & -1-\lambda \end{vmatrix} = 0 \Rightarrow (2-\lambda)(3-\lambda)(-1-\lambda) - 4 - 4 + 2(3-\lambda) + 4(2-\lambda) + 2(1+\lambda) = 0$$

$$\Rightarrow (6-2\lambda-3\lambda+\lambda^2)(-1-\lambda) - 8 + 6 - 2\lambda + 8 - 4\lambda + 2 + 2\lambda = 0$$

$$\Rightarrow (6-5\lambda+\lambda^2)(-1-\lambda) + 8 - 4\lambda = 0 \Rightarrow$$

$$\Rightarrow -6 - 6\lambda + 5\lambda + 5\lambda^2 - \lambda^2 - \lambda^3 + 8 - 4\lambda = 0$$

$$\Rightarrow -\lambda^3 + 4\lambda^2 - 5\lambda + 2 = 0 \Rightarrow (\lambda-1)^2(-\lambda+2) = 0$$

	λ^3	λ^2	λ	λ^0	
	-1	4	-5	2	
1	-1	3	-2	0	
1	-1	2	0		

$$\Rightarrow -(\lambda-1)^2(\lambda-2) = 0$$

$$\Rightarrow \lambda_1 = 1 \Rightarrow a(\lambda_1) = 2$$

$$\lambda_2 = 2 \Rightarrow a(\lambda_2) = 1$$

$$\lambda_1 = 1 \Rightarrow Av = \lambda_1 v \Rightarrow \begin{pmatrix} 2 & 1 & -2 \\ 2 & 3 & -4 \\ 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 1 & -2 \\ 2 & 2 & -4 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} x+y-2z=0 \\ 2x+2y-2z=0 \\ x+y-2z=0 \end{cases} \Rightarrow$$

$$\Rightarrow x = -y + 2z \Rightarrow V_{\lambda_1} = \left\{ \begin{pmatrix} -y+2z \\ y \\ z \end{pmatrix} \mid y, z \in \mathbb{R} \right\}$$

$$V_{\lambda_1} = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ deoarece } V_{\lambda_1} = 2 \Rightarrow g(\lambda_1) = 2$$

$$\lambda_2 = 2 \Rightarrow Av = \lambda_2 v \Rightarrow \begin{pmatrix} 0 & 1 & -2 \\ 2 & 1 & -4 \\ 1 & 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} y-2z=0 \Rightarrow y=2z \\ 2x+y-4z=0 \\ x+y-3z=0 \end{cases} \quad \underline{\hspace{1cm}} \quad x=3z-2z \Rightarrow x=z$$

$$V_{\lambda_2} = \left\{ \begin{pmatrix} z \\ 2z \\ z \end{pmatrix} \mid z \in \mathbb{R} \right\} = \text{Sp} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\} \quad \dim V_{\lambda_2} = 1$$

$$g(\lambda_2) = 1$$

$$\begin{array}{l} a(\lambda_1) = g(\lambda_1) = 2 \\ a(\lambda_2) = g(\lambda_2) = 1 \end{array} \Rightarrow \text{matricea } A \text{ e diagonalizabilă}$$

$$\Rightarrow A = P \cdot D \cdot P^{-1} \text{ unde } P = \begin{pmatrix} -1 & 2 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$P^t = \begin{pmatrix} -1 & 1 & 0 \\ 2 & 0 & 1 \\ 1 & 2 & 1 \end{pmatrix} \quad P^{-1} = \begin{pmatrix} -2 & -1 & 4 \\ -1 & -1 & 3 \\ 1 & 1 & -2 \end{pmatrix}$$

$$A = \begin{pmatrix} -1 & 1 & 0 \\ 2 & 0 & 1 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & -1 & 4 \\ -1 & -1 & 3 \\ 1 & 1 & -2 \end{pmatrix}$$